Research Statement

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I am a macroeconomist working on economic development. I study how the heterogeneous behavior of individual households, firms, and farms responding to their unique constraints in the presence of frictions shapes aggregate growth. My approach combines quantitative heterogeneous-agent models with experimental and observational micro-level data. I study the idiosyncrasies in the observed behavior of individual firms and households, which motivate and discipline the development of theory. I then construct a model to understand this behavior and trace out its quantitative impact on the aggregate economy.

My agenda has developed along three lines, each motivated by a large and stubborn gap between low-income and developed countries:

1. *Migration frictions and urban–rural income gaps*. Why do aggregate urban–rural income gaps persist in low-income countries, and what effect do policies targeting migration have on the aggregate economy? This line is motivated by the large urban–rural income gaps in low-income countries (Lagakos 2020).

2. *Firm heterogeneity, misallocation, and aggregate productivity*. How do input market frictions distort the decisions of individual firms and farms in the presence of heterogeneous technologies, and what effect do these frictions have on the aggregate output of an economy? This line is motivated by the large gaps in measured misallocation between low-income and developed countries (Hsieh and Klenow 2009).

3. *Subsistence farming and aggregate productivity*. How does the nutrition-driven behavior of subsistence farmers aggregate to determine national agricultural productivity, and what effect are policies targeting smallholder farmers likely to have on the aggregate economy? This line is motivated by the large agricultural productivity gaps between low-income and developed countries, and between agricultural and non-agricultural sectors of low-income countries (Gollin, Lagakos, and Waugh 2014).

Migration Frictions and Urban-Rural Income Gaps

Income gaps between rural and urban areas are large and persistent in many low-income countries (Lagakos 2020). A growing experimental literature identifies a number of migration frictions that contribute to these gaps, but has little to say on their relative importance and aggregate implications. The first area of my research agenda focuses on bridging the gap between experimental evidence and macroeconomic implications.

In **"The Macroeconomic Effects of Lifting Barriers to Migration"** (conditionally accepted at the *Journal of Monetary Economics*, with Travis Baseler), we build and estimate a dynamic general-equilibrium model of rural–urban migration which nests several of the key migration frictions emphasized in the literature. We discipline the model using large-scale data from Kenya—the site of several recent migration experiments trialling cash transfers, information interventions, and unemployment insurance. We reproduce each experiment within the model, then use it to conduct three sets of counterfactual quantitative exercises. First, we evaluate the long-run effects of each intervention: migration and income treatment effects grow substantially beyond the experimental endlines. Second, we evaluate the macroeconomic effects of scaling the interventions: unemployment insurance and information treatments modestly boost urbanization and GDP if rolled out nationally. Third, we assess the aggregate importance of different frictions. The model identifies information frictions as the most important but shows that most of the three-fold urban–rural income gap is left unexplained by any friction studied in recent experimental literature: the biggest obstacles to migration remain elusive. I have presented this work at the Carnegie-Rochester-NYU Public Policy Conference, NEUDC, and RIDGE forum (scheduled).

In **"Reaching the Novice or Nudging the Expert? Information Delivery and the Returns to Migration"** (under review, with Zachary Barnett-Howell, Travis Baseler, and Thomas Ginn), we dig deeper into information interventions, assessing their effects experimentally and evaluating their macroeconomic impacts quantitatively. We study information dissemination in the context of rural-urban migration, where spatial income gaps are large and the returns to moving are heterogeneous. In a large-scale, cluster-randomized experiment in rural Kenya, we test three delivery modes: providing information about the capital city privately to rural households, facilitating private conversations between

rural households and urban workers, and broadcasting information publicly in villages. Each mode increases migration by similar amounts. Yet both private modes generate significant economic gains, while public dissemination does not. We find that private delivery influences those with less migration experience—who exhibit higher income gains at the margin—compared to public dissemination. We build and estimate a model of constrained migration to rationalize these patterns and quantify the role of selection on experience in explaining the economic returns across delivery modes. Our results show that the value of information depends not only on whether it changes behavior but also on which households respond. I have presented this work at SED, STEG, NEUDC, NTxEC (where it won the "Best Paper Award, Junior Economist"), and the Kiel-CEPR-World Bank AEDC (scheduled).

My future work in this direction is suggested by the open-ended result of *The Macroeconomic Effects of Lifting Barriers to Migration*. First, the validation of the model with RCT evidence highlighted two mechanisms that would help the model match experimental evidence better: risk aversion heterogeneity and local demand spillovers via economic slack (such as low capacity utilization widespread in low-income country firms). Second, the model suggests that most of the threefold urban–rural income gap is down to the "residual migration cost": allowing for heterogeneity in this residual and identifying household characteristics correlated with it would hint at the migration frictions the field should study. I plan to build on my quantitative framework to incorporate both, which will make further progress in our understanding of the sources of urban–rural income gaps and limited urbanization in low-income countries.

Firm Heterogeneity, Misallocation, and Aggregate Productivity

The second area of my research agenda is studying how the heterogeneous behavior of individual firms interacts with distortionary frictions that harm the aggregate allocation of inputs. The misallocation of production inputs between firms has been shown to be a big driver of cross-country income differences (Hsieh and Klenow 2009): it is especially severe between *farms* within the agricultural sectors of low-income countries (Chen, Restuccia, and Santaeulàlia-Llopis 2023). My work contributes by studying firms' heterogeneous choice of products and production technologies, which significantly affects the propagation of misallocative frictions in the economy—and the measurement of what constitutes

a misallocative friction and what does not.

In **"Misallocation and Product Choice"** (under review, with Sudhir Singh), we study the costs of misallocation of inputs between multi-product firms that endogenously choose among heterogeneous products. Existing estimates of the costs of misallocation of inputs have relied on models of single-product firms using a single aggregate production function. Using rich farm-crop-level data from India, we estimate product-level production functions and find that they are meaningfully different from one another and from the aggregate one. We build a general equilibrium model of firm-level misallocation in which multi-product firms (or farms) are able to choose the set and mix of heterogeneous products. Equalizing cross-farm input-market frictions raises aggregate productivity by 137% in the multi-product model and 167% in a conventional single-product model: the single-product model overstates the aggregate productivity cost of misallocation between Indian farms by 22%. I have presented this work at Midwest Macro, SED, SMU, and NEUDC.

In **"Input-Driven Production Technology Heterogeneity and the Allocation of Inputs"** (working paper, with Sudhir Singh), we study whether heterogeneity in the intrinsic features of production inputs may generate heterogeneity in production technologies optimally chosen by firms, leading existing estimates of misallocation to overstate its costs. We use a state-of-the-art clustering algorithm to group Indian agricultural plots into land types by the intrinsic physical features of each plot. We find that production functions are significantly heterogeneous across land types, which we confirm with a placebo-like randomization inference test: some types of land are better suited to land-intensive technology, others to labor-intensive technology, and so on. We build a model in which heterogeneous farmers face distortions and rent land of several types. We will use the model to quantify the cost of misallocation for India's aggregate agricultural productivity and compare it to conventional models that assume homogeneous production technology.

My future work in this direction aims to study input-market frictions, domestic trade costs, and international trade costs jointly. Input market distortions generating misallocation across farmers and intra-national trade costs preventing specialization by farmers have been separately explored as important drivers of low agricultural productivity, but their interaction and relative importance have not been studied. In **"Farms, Trade, and Misallocation"** (a work in progress, with Sudhir Singh), we develop an empirical and theoretical strategy for understanding agricultural misallocation and intra-national trade

costs jointly. First, we use rich farm-level survey data from India to document the effect of distorted input markets and poor market access on the behavior of India's farmers and their (in)ability to specialize and take advantage of intra-national and international trade. Second, we develop a novel theoretical framework that embeds misallocation in a model of spatial trade with heterogeneous farmers, land, and crops. Next, we will use the model to quantify the relative contributions of input market frictions, intra-national trade costs, and international trade barriers to keeping the agricultural productivity of India's farmers low. We will simulate alternative policies and identify the ones most effective at alleviating the costliest frictions that constrain India's farmers and accelerating the structural transformation of low-income countries like India.

Subsistence Farming and Aggregate Productivity

The third area of my research agenda is studying the link between the behavior of subsistence farmers and aggregate agricultural productivity. Many low-income countries are still dominated by agriculture, which tends to be extremely unproductive and done mainly by semi-subsistence smallholders. My work explicitly brings farmers' nutritional needs and heterogeneous product choice into the picture, showing that considering these dimensions is crucial for understanding the behavior of subsistence farmers and for tracing out its aggregate implications.

In **"Nutrition Demand, Subsistence Farming, and Agricultural Productivity"** (*Journal of Development Economics*, 2026), I propose nutrition demand as a mechanism that can explain what subsistence farmers in low-income countries choose to consume and grow on their land, and that ultimately contributes to keeping the aggregate agricultural productivity low. In the first contribution, I exploit rich household-level data to document a set of novel facts on how the products these farm-operating households choose to eat and the products they choose to grow depend on the scale of the farm. The smallest farmers specialize their consumption and subsistence production in obtaining calories, medium farmers diversify both consumption and subsistence production, and large farmers commercialize the farm. In the second contribution, I build a model in which farmers have explicit dietary energy needs. I find that the interaction of this novel nutritional channel with domestic trade costs can explain the new empirical micro-level facts. In the third contribution, because subsistence farmers have to use their farms to target their nutri-

tional needs rather than their comparative advantage, the model suggests that allowing farmers even partially to leave subsistence and commercialize can raise the productivity of the agricultural sector by over 40% at the aggregate level. This paper was funded by a STEG small research grant. I have presented this work at SED and CSAE, among others.

My future work in this direction will use this framework for policy analysis. In **"The Macroeconomic Effects of Nutrition-Sensitive Policies"**, a follow-up project that is in the preliminary stage, I intend to apply the model developed in the paper above to analyze the nutrition-sensitive policies targeting smallholder farmers, which are at the center of welfare programs in many low-income countries, including Malawi. Because the model I developed captures the incentives for calorie-oriented staple crop cultivation, variety-oriented crop diversification, and market-oriented cash crop production, it is well-suited for analyzing the economic and, crucially, the nutritional effects of many existing and proposed policies. It can also predict how the impact of the policy on a particular type of farmers depends on their characteristics and the properties of the targeted crop. Finally, the model allows for aggregate counterfactuals that predict the general-equilibrium effects of nutrition-sensitive policies on the entire agricultural sector and the entire economy.

References

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